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Microwave Engineering Theory

Project Report Submission

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Question Details:

Design an air-filled E-plane Tee for WR 62 waveguide. Utilizing Electric Field distributions and S-parameter plots, demonstrate its characteristics as an equal power divider and explain why the two output signals are in phase opposition (180° out of phase).

Circuit Diagram/Structure:

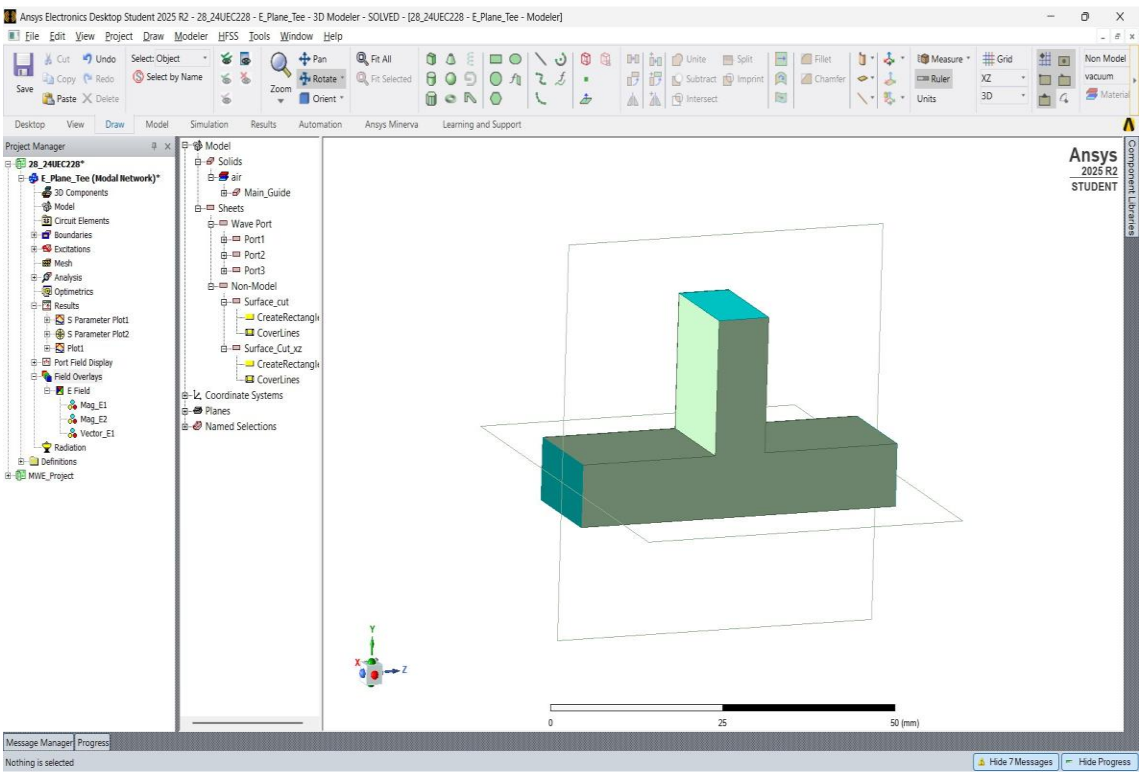

Fig.1: E Plane Tee Design in HFSS using the dimensions of WR62 Waveguide

The E-Plane Tee is a three-port waveguide junction formed by connecting an auxiliary waveguide to the centre of the broad wall of the main waveguide. It is called an E-plane tee because the side arm lies in the plane of the electric field of the dominant TE_{10} mode.

The structure was designed using WR-62 rectangular waveguide dimensions in ANSYS HFSS. The waveguide dimensions used are:

- Width (a) = 15.8 mm
- Height (b) = 7.9 mm

The model consists of:

- Port 1 and Port 2 forming the main waveguide
- Port 3 connected perpendicular to the main guide
- Perfect electric conductor (PEC) boundaries
- Wave ports assigned at all three ports

The simulation was performed over the microwave frequency range to observe transmission, reflection, and field distribution characteristics.

For WR-62:

$$a = 15.8 \text{ mm}$$

Cutoff frequency for TE_{10}:

$$f_c = \frac{c}{2a}$$

Substituting values:

$$f_c = \frac{3 \times 10^8}{2 \times 15.8 \times 10^{-3}}$$
$$f_c \approx 9.49 \text{ GHz}$$

The simulated cutoff frequency closely matches the theoretical cutoff frequency of approximately 9.49 GHz for the WR-62 waveguide.

WR-62 waveguide operates primarily in the Ku-band frequency range (12–18 GHz), and the simulation at 15 GHz lies within this operating region.

Simulation Results:

S Parameter Plot Showing Equal Power Distribution in Port1 and Port2 when E-arm is Excited:

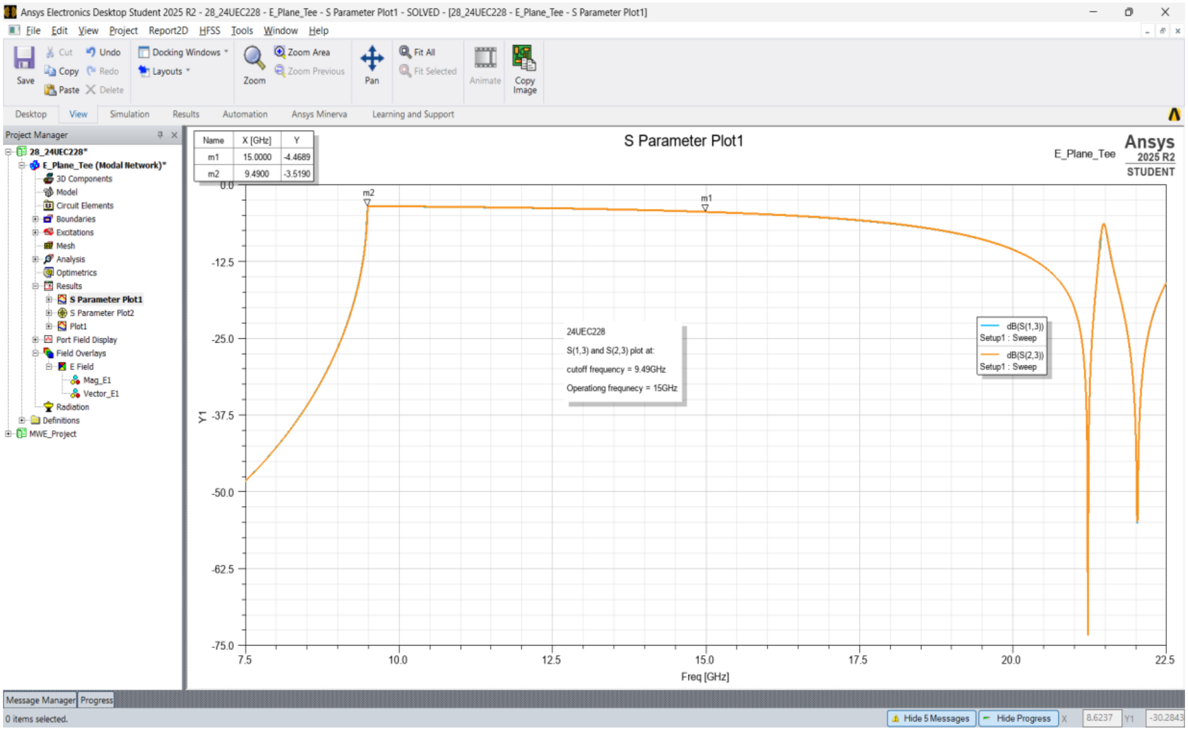

Fig.2: Plot of S_{13} and S_{23} v/s Frequency with markers at $f = f_c$ and f = 15GHZ

S Parameter Plot Showing 180° Phase Difference:

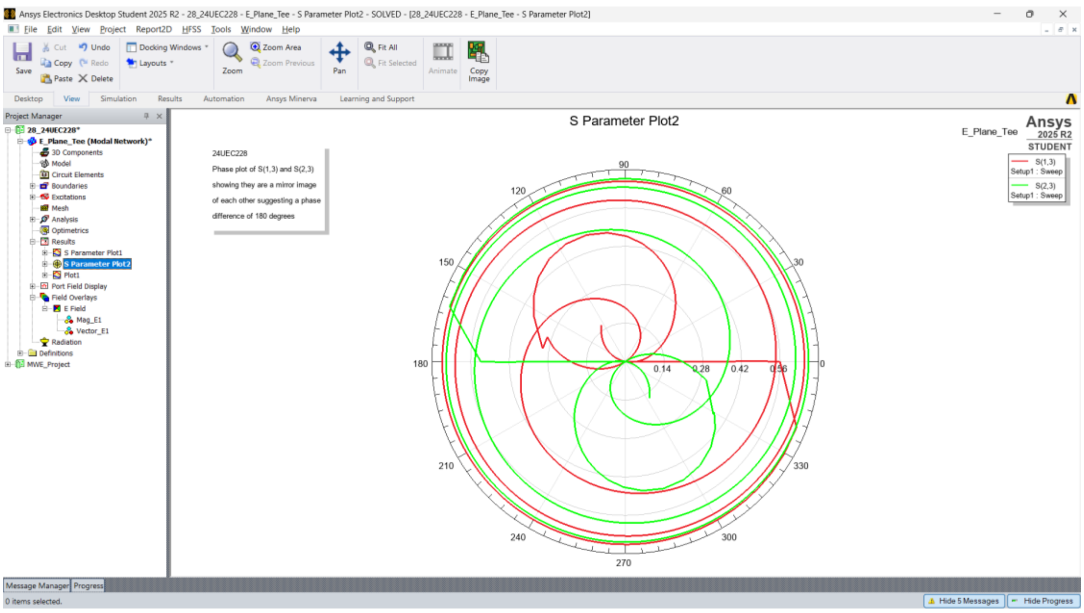

Fig.3: S_{13} and S_{23} Polar Plot Showing 180° Phase Difference (mirror images)

S Parameter Plot with phase difference clearly plotted:

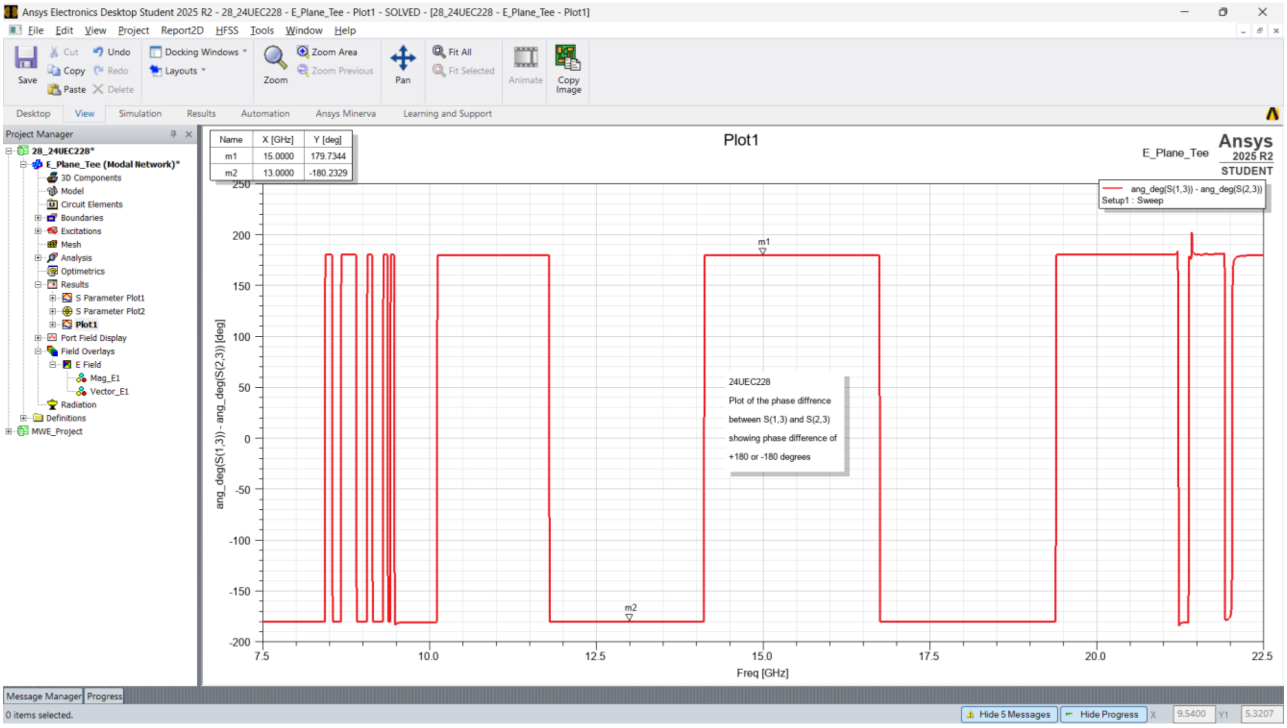

Fig.4: Rectangular Plot Showing ang(S_{13}) – (S_{23}) = |180°|

Magnitude Distribution of E-Field on Surface Cut Plane (YZ):

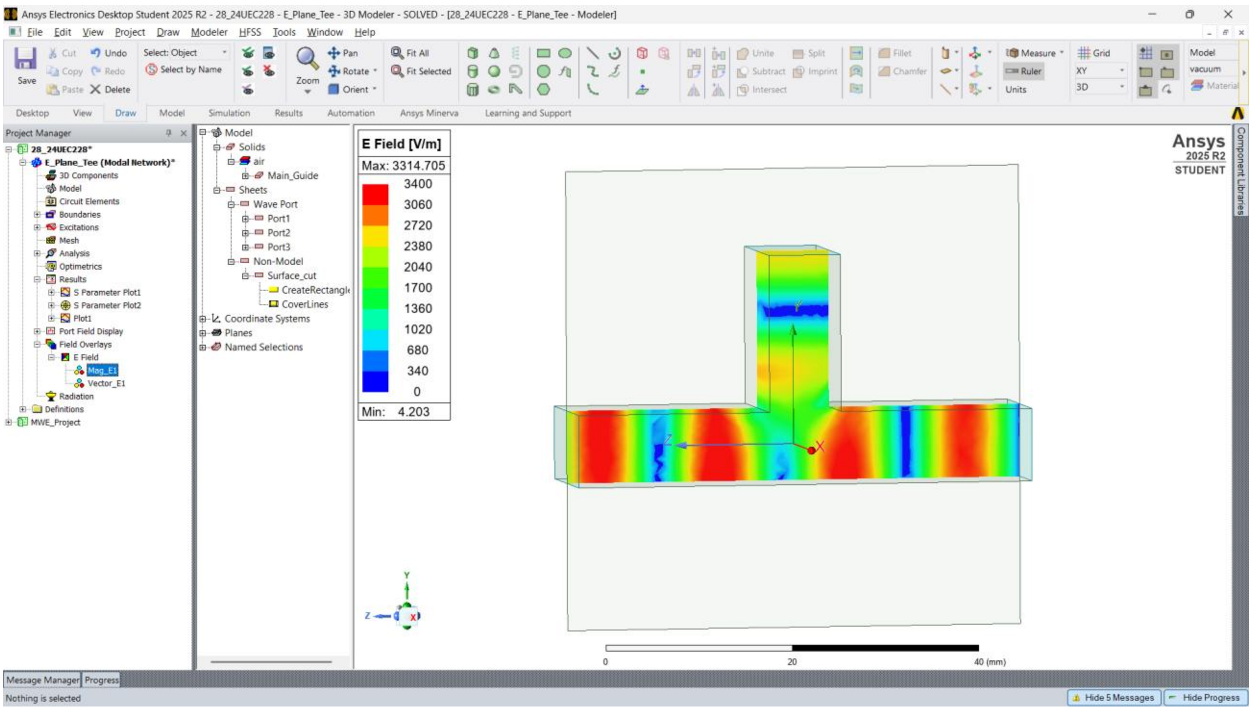

Fig.5: Mag_E Plot Showing Equal Power Distribution in Port1 and 2 at f = 15GHz and Phase = 60°

Magnitude Distribution of E-Field on Surface Cut Plane (XZ) showing TE_{10} Mode:

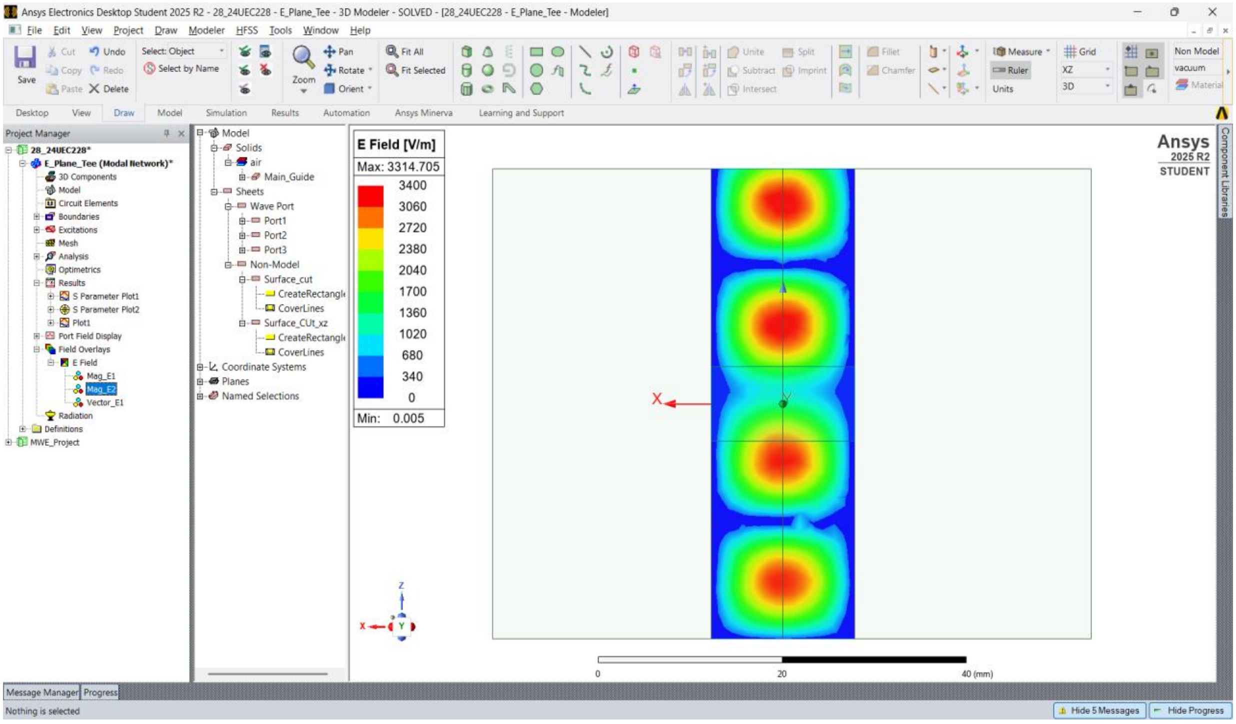

Fig.6: Mag_E Plot on XZ Surface Cut Plane Showing Standing Wave Pattern for TE_{10} Mode of Propagation

Vector E-Field on Surface Cut Plane (YZ) Showing 180° Phase Difference:

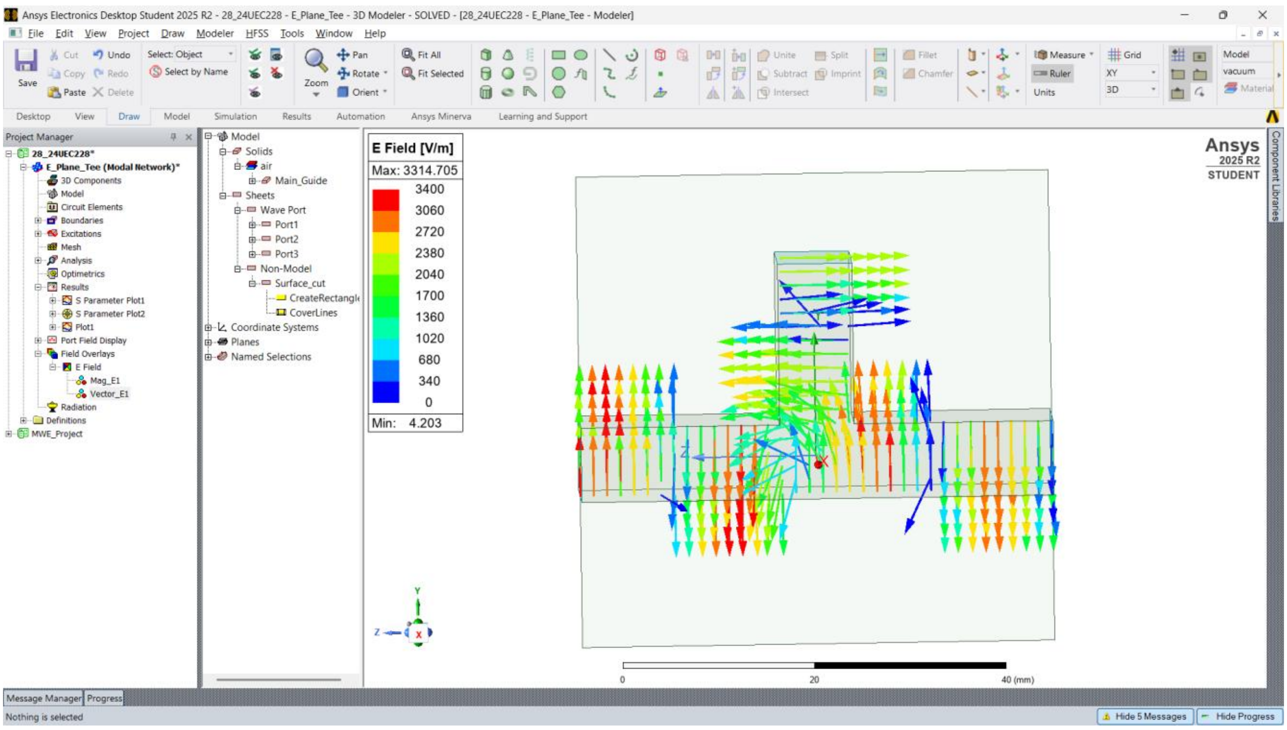

Fig.7: Vector_E Plot showing 180° phase Difference at Port 1 and 2 at f = 15GHz and Phase = 60°

Analysis of Result:

TE_{10} Mode Verification: TE_{10} is the dominant mode because it has the lowest cutoff frequency among all possible modes in a rectangular waveguide. The XZ cross-sectional plot (Fig. 6) shows a sinusoidal E-field variation across the broad wall, with maximum field at the centre and nulls at the sidewalls. This confirms propagation of the dominant TE_{10} mode in all three arms of the junction, with no evidence of higher-order mode excitation within the operating band.

Equal Power Division: From the S-parameter magnitude plot (Fig. 2), dB (S (1,3)) = dB (S (2,3)) = −4.46 dB at 15 GHz. The two curves overlap across the entire passband (9.5–20 GHz), confirming equal power division to both output ports. In an ideal lossless equal power divider, each output should be −3 dB. However, the obtained value of approximately −4.46 dB occurs due to impedance mismatch and reflections at the junction of the unmatched E-plane tee.

Phase Opposition Explanation: When a signal is fed into the E-arm (Port 3), it excites the TE_{10} mode in the main guide. At the junction, the signal splits into two waves travelling in opposite directions toward Port 1 and Port 2. Since the E-field of the TE_{10} mode is oriented along the Y-axis and the two waves travel in +Z and −Z directions respectively, the boundary condition at the junction requires the transverse E-field to reverse direction for one of the output arms. This results in the two output signals being 180° out of phase, as confirmed by the phase difference plot (Fig. 4), which shows ≈ +180.14° at 15 GHz and ≈ −179.88° at 13 GHz.

Vector Field Confirmation: The E-field vector plot (Fig. 7) provides direct visual evidence of the phase opposition. The vectors in the left arm (Port 1) point upward (+Y) while those in the right arm (Port 2) point downward (−Y) at the same instant in time, which is a spatial representation of the 180° phase difference. This behaviour is the defining characteristic that distinguishes the E-plane Tee from the H-plane Tee, where both outputs are in phase.

Standing Waves and Mismatch: The standing wave pattern visible in Fig. 5 and Fig. 6 results from the superposition of the incident and reflected waves at the unmatched junction. Multiple field maxima and minima are visible along the E-arm, which is consistent with the poor return loss expected at Port 3 for an unmatched E-plane Tee.

Conclusion:

The air-filled E-plane Tee for the WR-62 waveguide was successfully designed and simulated using Ansys HFSS 2025 R2. Simulation results confirm that the structure operates as an equal power divider, with S (1,3) = S (2,3) = −4.46 dB at 15 GHz across the Ku-band. The phase difference between the two output ports was verified to be approximately 180° throughout the operating bandwidth, consistent with the theoretical behaviour of an E-plane junction. The E-field distribution plots confirm pure TE_{10} mode propagation in all arms and provide direct visual evidence of the anti-phase field relationship between the two collinear output ports. The small excess insertion loss beyond the ideal −3 dB is attributed to junction mismatch, which is an inherent property of the unmatched E-plane Tee.